Higher National Diploma in Civil Engineering
4th Year, First Semester Examination- 2021
CE3225 –Buildings and Structures

Instructions for Candidate:

Answer any five (05) questions only
All question carries equal marks

No. of Questions : 06
No. of Pages : 02
Time : 03 (hours)

1.

I. "A landfill or land disposal is an integral part of a sound solid waste management strategy" Rationalize this statement. (04 Marks)

II. What basic aspects must you consider while selecting a site for landfill implementation? (05 Marks)

III. Illustrate the components of an ideal septic tank for a residence meant for 4-6 persons with the aid of a neat sketch. (05 Marks)

IV. List down 4 types of hot water systems and discuss the advantages and disadvantages of each system. (06 Marks)

(Total 20 Marks)

2.

I. Briefly explain the active and passive techniques used to control the temperature within the multistory building. (05 Marks)

II. Explain the operating principle behind the *air conditioner* (Use sketches for the explanation and highlight the key parts) (08 Marks)

III. Discuss the different types of artificial lighting with examples (04 Marks)

IV. What are the key points that you must consider for the energy-efficient lighting design? (03 Marks)

(Total 20 Marks)

3.

I. Explain the fire safety measure that should be considered in good building design (06 Marks)

II. Briefly describe the fire regulation in Sri Lanka. (04 Marks)

III. Briefly explain the following active fire protection systems

a) Automatic Fire Extinguishers
b) Hose Reels System
c) Fire sprinkler system

d) Foam Fire Suppression System

(08 Marks)

IV. Why smoke extraction & ventilation is important during the fire incident? (02 Marks)

(Total 20 Marks)

4.

I. Briefly explain the vertical circulation in a multistory building with examples.

(04 Marks)

II. Describe the "preliminary planning clearance" in building approval process.

(04 Marks)

III. Give short description to following terms use in building approval process in Sri Lanka.

a) Environmental Impact Assessment (EIA)
b) Traffic Impact Assessment (TIA)
c) Council Drawing
d) Building regulation in Sri Lanka

(08 Marks)

IV. State types of staircase and components of a staircase. (04 Marks)

(Total 20 Marks)

5.

I. Name 3 organizations that provide green building certification which is valid in Sri Lanka (03 Marks)

II. List down the main goals of the Green Rating System? (06 Marks)

III. How does "sustainable site selection" contributes to the Green building concept?

(05 Marks)

IV. Discuss the advantages of green buildings over the conventional building.

(06 Marks)

(Total 20 Marks)

6.

I. Name 3 skyscrapers in Sri Lanka (03 Marks)

II. Discuss the advantage of skyscrapers construction in urban areas of a country

(04 Marks)

III. What are the safety aspects that should be considered in skyscraper design?

(04 Marks)

IV. List down 4 structural systems that utilize for resisting the lateral loads in high rise building construction. (04 marks)

V. What are the approvals that should be obtained prior to construction of a skyscraper in Sri Lanka? (05 marks)

(Total 20 Marks)